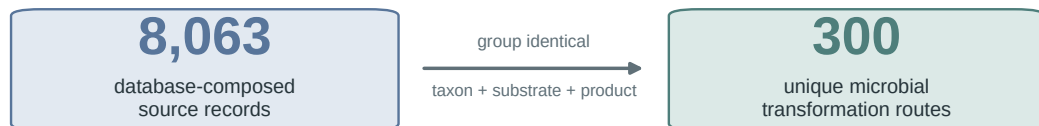


a Database records contract after route-level deduplication

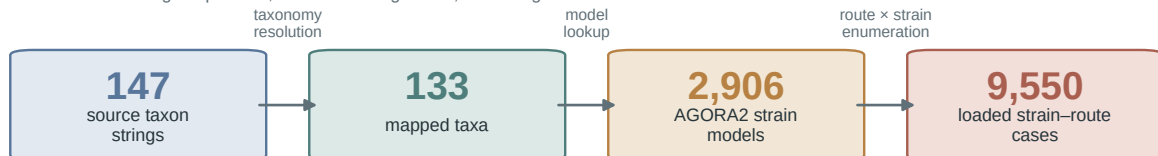
Counting unit changes from composed source records to unique taxon-substrate-product routes.



Deduplication is a contraction; box area is not proportional to count.

b Taxonomic mapping expands routes into strain-model evaluations

This is a coverage expansion, not a screening funnel; each stage uses a different unit.



133/147 taxa mapped; 21 no-model route placeholders retained (9,571 matrix rows total).

c Chemical and flux criteria leave a small model-supported subset

Counts remain strain-route combinations until the final collapse to unique routes.



One *E. coli* tryptophan → indole route contributed 987/1,058 F1-positive strain cases (93.3%); route counts are the primary summary.

F1 required eligible taxonomy, verified structures, native exchange reactions and increased product export under curated anaerobic model conditions.